

# Acquiescence Is Not Personality: Why Human Psychometric Instruments Fail on LLMs

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## Abstract

A growing literature treats LLMs as personality-bearing agents, administering human questionnaires and reporting scores. We ask the prior question: do these instruments measure anything at all when applied to LLMs? We gave a 221-item battery (IPIP-NEO-120, SD3, ZKPQ-50-CC, EPQR-A) to 20 LLMs across 8 families under 17 persona conditions (375,700 responses) and ran six standard psychometric validation checks: internal consistency, factor structure, reverse-item consistency, convergent validity, measurement invariance, and variance decomposition. Every check failed, and every failure traces back to one root cause: acquiescence—a blanket tendency to agree with statements regardless of direction, produced by safety training. Because LLMs agree with both forward-coded and reverse-coded items, Cronbach’s  $\alpha$  tracks reverse-item density perfectly ( $\rho = -1.0$ ): domains with few reverse items (Extraversion,  $\alpha = 0.93$ ) pass while domains with many (Agreeableness,  $\alpha = 0.06$ ) collapse. Because acquiescence inflates correlations across all domains—making distinct traits appear similar—factor analysis collapses the expected five-factor structure into just three factors, in every model and every family. Inter-model variance falls below 1%, rendering cross-model comparisons meaningless. Persona prompts shift scores by over one SD but do not restore validity. These instruments do not measure personality—they capture a uniform agreeable response style imposed by safety training.

## 1 Introduction

A growing literature administers human personality questionnaires to LLMs, computes scores, and reports personality profiles (Miotto et al., 2023; Bhandari et al., 2025; Serapio-García et al., 2025; Sorokovikova et al., 2023; Lee et al., 2025). Some propose using LLMs as “silicon samples” to simulate survey respondents (Horton, 2023). None

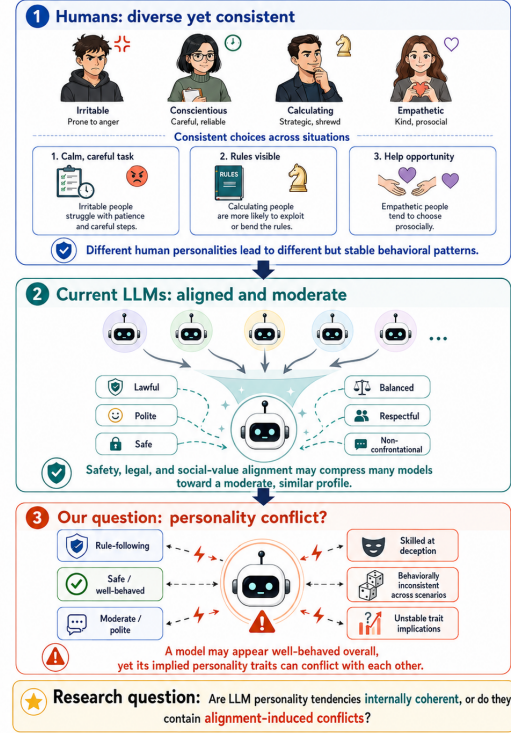


Figure 1: The question this paper asks. Rather than reporting LLM personality scores, we test whether the instruments produce valid measurement at all.

of this work asks the prior question: *do these instruments actually measure anything meaningful in LLMs?*

A personality questionnaire is only valid for a population if it passes basic psychometric checks: internal consistency (Cronbach’s  $\alpha$ ), correct factor structure, convergent validity, and measurement invariance across subgroups (Cronbach, 1951; Lord and Novick, 1968). In human samples, these checks are standard practice before any instrument is used. For LLMs, they are almost never performed. The scores are reported as if they reflect genuine traits, with no evidence that the measurement foundation holds.

The cracks are already visible. Sühr et al. (2025)

found that LLMs endorse both forward-coded and reverse-coded items simultaneously. Tjuaatja et al. (2023) showed systematic response biases in survey settings. Acerbi and Stubbersfield (2024) and Salecha and Ireland (2024) found that LLMs shift responses toward socially desirable endpoints. Pelet et al. (2025) showed that reordering questions produces different scores. Huang et al. (2024) found that scale reliability is sensitive to instructions. Each finding identifies a symptom. None identifies the root cause.

We do. We administered a 221-item battery (IPIP-NEO-120, SD3, ZKPQ-50-CC, EPQR-A) to 20 LLMs across 17 role-conditioned prompts and ran every standard psychometric validation check on the resulting 375,700 responses. Every check failed, and every failure traces back to a single mechanism: *acquiescence*—a blanket tendency to agree with statements regardless of direction, produced by safety training. The evidence forms one causal chain. At the item level, LLMs agree with both forward and reverse items, so internal consistency ( $\alpha$ ) correlates perfectly with reverse-item density ( $\rho = -1.0$ ). At the domain level, the same bias distorts inter-domain correlations so that exploratory factor analysis recovers 3 factors instead of 5. At the model level, safety training pushes every model toward the same agreeable response style, so inter-model variance falls below 1%. Persona prompts shift scores but do not restore measurement validity.

## 2 Related Work

### 2.1 Personality Measurement in LLMs

Researchers have administered personality questionnaires to LLMs and reported personality profiles (Miotto et al., 2023; Serapio-García et al., 2025; Bhandari et al., 2025; Sorokovikova et al., 2023; Lee et al., 2025; Hagendorff, 2024). But almost none of these studies check whether the instruments actually work on LLMs the way they work on humans. Benchmarks show systematic deviations between LLM and human responses (Xie et al., 2025), temporal stability is limited (Bodroža et al., 2024), and measurement invariance is assumed rather than tested (Sühr et al., 2025). Chen et al. (2025) noted that most existing frameworks skip psychometric validation entirely.

**Our contribution.** Previous work asks “what personality does this LLM have?” and reports scores. We ask a different question: do these instruments

measure anything meaningful at all when given to LLMs? Instead of profiling, we run the full set of standard psychometric validation checks (internal consistency, factor structure, convergent validity, reverse-item consistency, measurement invariance, and variance decomposition) on a single dataset covering 20 models and 221 items.

### 2.2 Response Style and Acquiescence Bias

Survey respondents do not always answer based on content alone. Acquiescence bias (agreeing with statements regardless of direction) is one of the most studied response styles (Paulhus, 1991; Lord and Novick, 1968; Baumgartner and Steenkamp, 2007). It is especially damaging for instruments that rely on reverse-coded items. Acerbi and Stubbersfield (2024) and Salecha and Ireland (2024) showed that LLMs also exhibit social desirability biases in Big Five surveys, and Responsible NLP Group (2025) confirmed that acquiescence in LLMs is robust across languages.

**Our contribution.** These studies documented that acquiescence and social desirability exist in LLM responses. Our approach is different: we quantify the measurement damage they cause by decomposing the bias at item level, separating forward-coded and reverse-coded agree rates, and comparing across instruments with different content (normal personality vs. dark traits).

### 2.3 Alignment Effects on LLM Behavior

RLHF and safety training make LLMs more helpful and harmless, but also more homogeneous (Ouyang et al., 2022; Bai et al., 2022). Sharma et al. (2024) showed that aligned models tend to echo user opinions even when wrong, and Zheng et al. (2025) linked Agreeableness scores to sycophancy in role-playing LLMs. Wang et al. (2025) argued that RLHF encodes a specific notion of acceptability that may override genuine variation.

**Our contribution.** These studies examined alignment effects on individual behaviors. We take a different angle: we test whether alignment produces a universal response style across model families via DIF analysis, and we quantify how far persona injection can push a model from its default behavior (Persona Separation Degree).

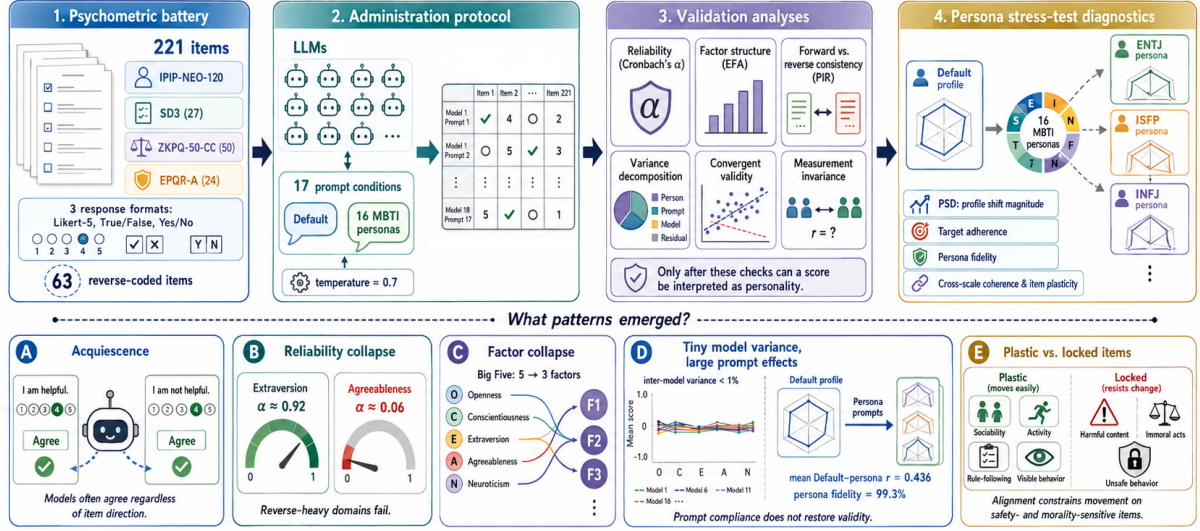


Figure 2: Overview of the methodology. We administer a multi-instrument psychometric battery to LLMs, normalize the response matrix, and run standard validation checks before interpreting any score as evidence of personality.

Table 1: Psychometric instruments in the 221-item battery.

| Instrument   | Items      | Format           | Domains   | Rev.      |
|--------------|------------|------------------|-----------|-----------|
| IPIP-NEO-120 | 120        | Likert-5         | 5         | 41        |
| SD3          | 27         | Likert-5         | 3         | 5         |
| ZKPQ-50-CC   | 50         | T/F              | 5         | 12        |
| EPQR-A       | 24         | Y/N              | 4         | 5         |
| <b>Total</b> | <b>221</b> | <b>3 formats</b> | <b>17</b> | <b>63</b> |

Table 2: All 20 tested LLMs grouped by model family.

| Family    | Models                                                              |
|-----------|---------------------------------------------------------------------|
| Anthropic | Claude Opus 4.6, Sonnet 4.6                                         |
| OpenAI    | GPT-5.5, GPT-5.2                                                    |
| DeepSeek  | V3.2, V4-Flash, V4-Pro                                              |
| Google    | Gemini-3-Flash, Gemini-3.1-Flash-Lite, Gemini-3.1-Pro, Gemini-3-Pro |
| Alibaba   | Qwen3-235B-A22B, Qwen3.5-122B-A10B, Qwen3.5-397B-A17B               |
| Moonshot  | Kimi-K2.5, Kimi-K2.6                                                |
| MiniMax   | M2.7                                                                |
| Zhipu     | GLM-4.6V, GLM-4.7, GLM-5.1                                          |

### 3 Methodology

#### 3.1 Psychometric Instruments

We administered a 221-item battery comprising four validated instruments (Table 1): IPIP-NEO-120 (Johnson, 2014), SD3 (Jones and Paulhus, 2014), ZKPQ-50-CC (Aluja et al., 2006), and EPQR-A (Francis et al., 1992). Together they cover 17 domains across three response formats (Likert-5, True/False, Yes/No) and include 63 reverse-coded items.

#### 3.2 Models and Administration Protocol

We tested 20 LLMs spanning 8 model families (Table 2), covering both proprietary and open-weight systems.

Each model was tested under 17 conditions: a **Default** prompt (“Please answer the following personality questionnaire honestly.”) and 16 **MBTI persona** prompts, each assigning a specific MBTI type. All administrations used temperature = 0.7. Each (model, persona, item) cell was sampled 5 times ( $k=5$ ). Responses are averaged across sam-

ples within each cell. This yielded  $20 \times 17 \times 221 = 75,140$  item-level scores (averaged from 375,700 raw API calls).

#### 3.3 Analysis Pipeline

We applied six checks that psychometricians use before trusting any instrument on a new population.

**Internal consistency.** We compute Cronbach’s  $\alpha$  per IPIP domain for each model, using the 17 persona conditions as observations. For a domain with  $k$  items, let  $X_{pi}$  be the scored response to item  $i$  under persona  $p$ . Then:

$$\alpha = \frac{k}{k-1} \left( 1 - \frac{\sum_{i=1}^k \text{Var}(X_i)}{\text{Var}(\sum_{i=1}^k X_i)} \right) \quad (1)$$

We average  $\alpha$  across the 20 models. Human benchmarks come from Johnson (2014).

**Factor structure.** Let  $\mathbf{x}_j \in \mathbb{R}^{17}$  be the vector of domain scores for observation  $j$  ( $j = 1, \dots, 340$ ). We standardize each domain to zero mean and unit variance, then compute the correlation matrix  $\mathbf{R}$ . We retain factors via the Kaiser criterion ( $\lambda_i > 1$ ) and validate with parallel analysis: we permute each column of the data matrix independently, recompute eigenvalues, and repeat 500 times. A factor is retained only if its observed eigenvalue exceeds the 95th percentile of the permuted distribution. We extract factor loadings via eigendecomposition of  $\mathbf{R}$  followed by varimax rotation.

**Reverse-item consistency.** For each domain that contains both forward-coded (+) and reverse-coded (−) items, we compute the Pairwise Inconsistency Rate (PIR). For Likert items, a forward-reverse pair ( $f, r$ ) is inconsistent when both responses fall on the same side of the midpoint:

$$\text{PIR} = \frac{1}{|F||R|} \sum_{f \in F} \sum_{r \in R} \mathbf{1}[\text{sgn}(f-3) = \text{sgn}(r-3)] \quad (2)$$

where  $F$  and  $R$  are the sets of forward and reverse item responses, and the midpoint is 3 on the 1–5 scale. For binary items, inconsistency is  $f = r$ .

**Variance decomposition.** We normalize all responses to a common 0–1 scale: Likert scores via  $x' = (x - 1)/4$ , binary scores kept as-is. We then decompose the total sum of squares sequentially:

$$\begin{aligned} \text{SS}_{\text{total}} &= \sum_i (y_i - \bar{y})^2 \\ &= \text{SS}_{\text{model}} + \text{SS}_{\text{domain}} \\ &\quad + \text{SS}_{\text{persona}} + \text{SS}_{\text{item}} + \text{SS}_{\text{resid}} \end{aligned} \quad (3)$$

where each component groups by the corresponding factor and computes the squared deviation of group means from the grand mean. We run this separately for Likert and binary scales.

**Convergent validity.** For each of 8 pre-specified construct pairs (e.g., IPIP Neuroticism vs. ZKPQ Neuroticism-Anxiety), we compute Pearson  $r_p$  and Spearman  $r_s$  across the 20 model-level means. A pair passes if the correlation sign matches the theoretical expectation.

**Measurement invariance.** For each model, let  $\mathbf{d} \in \mathbb{R}^{120}$  be the vector of IPIP-NEO-120 parsed values under the Default persona, and  $\mathbf{m}_k$  the same vector under MBTI persona  $k$ . We compute:

$$r_k = \text{corr}(\mathbf{d}, \mathbf{m}_k), \quad k = 1, \dots, 16 \quad (4)$$

Strong invariance requires  $r_k > 0.8$  for all persona pairs. We report the mean  $r$  across all  $20 \times 16 = 320$  pairs.

**Persona separation degree (PSD).** We quantify how far a persona prompt pushes a model from its default profile. After z-scoring each domain across all  $20 \times 17 = 340$  observations, we define:

$$\text{PSD}(m, p) = \frac{1}{\sqrt{17}} \|\mathbf{z}_{m,p} - \mathbf{z}_{m,\text{Default}}\|_2 \quad (5)$$

where division by  $\sqrt{17}$  yields a per-dimension effect size in SD units.

**Persona steering diagnostics.** PSD measures distance from Default but not direction. We define two targets: an **empirical target** using the leave-one-model-out centroid difference  $\mathbf{t}_{m,p}^{\text{emp}} = \bar{\mathbf{z}}_{-m,p} - \bar{\mathbf{z}}_{-m,\text{Default}}$ , and a **theoretical target** mapping MBTI letters to expected domain shifts (e.g., E  $\rightarrow$  higher Extraversion, F  $\rightarrow$  higher Agreeableness). For each persona shift  $\Delta_{m,p} = \mathbf{z}_{m,p} - \mathbf{z}_{m,\text{Default}}$ , we measure target adherence ( $\cos(\Delta_{m,p}, \mathbf{t}_{m,p})$ ) and off-target leakage. We also classify each profile to the nearest leave-one-out persona centroid, yielding a fidelity confusion matrix.

**Cross-scale persona coherence and item plasticity.** For overlapping constructs measured by multiple instruments (e.g., Extraversion across IPIP/EPQR/ZKPQ), we sign-align each construct and compute a coherence score  $|\overline{\Delta}|/|\overline{\Delta}|$ , where 1 means all scales move in the same direction and 0 means they cancel. At the item level, we compute each item’s mean absolute shift from Default and the rate at which forward/reverse item pairs move in theoretically opposite directions.

## 4 Experiments and Results

### 4.1 Default Personality Profiles

Fig. 3 shows the IPIP-NEO-120 scores under the Default (no-persona) prompt. All 20 models cluster in the same corner: high Agreeableness, high Conscientiousness, low Neuroticism. The inter-model range is narrowest for Agreeableness (0.92 on a 1–5 scale) and widest for Neuroticism (2.10). Every model looks like a “nice, responsible, emotionally stable” person—whether it was made by Anthropic, OpenAI, Google, or DeepSeek. Is this a genuine personality, or an artifact of alignment training? The sections that follow trace the evidence from item to domain to factor to model, identifying a single root cause.

### 4.2 Internal Consistency

The first diagnostic is whether items within a domain agree with each other. Table 3 shows the

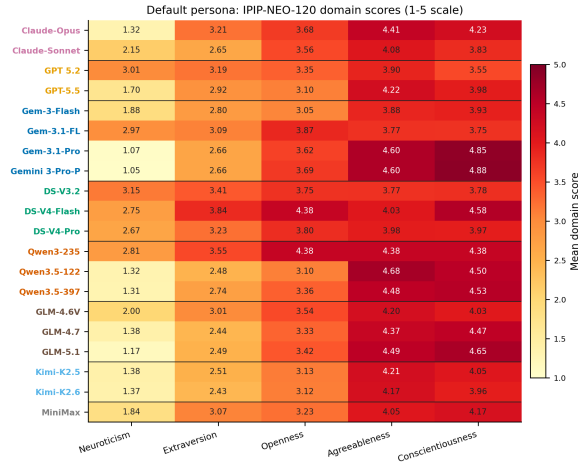


Figure 3: Default persona IPIP-NEO-120 domain scores (z-scored heatmap). All 20 models cluster in a “high A, high C, low N” region. Inter-model range is narrowest for Agreeableness (0.92 on a 1–5 scale) and widest for Neuroticism (2.10).

Table 3: Cronbach’s  $\alpha$  by IPIP domain: LLM vs. human norms. Mean $\pm$ SD across  $n=20$  models.

| Domain            | LLM $\alpha$      | Human $\alpha$ | Gap    |
|-------------------|-------------------|----------------|--------|
| Neuroticism       | 0.813 $\pm$ 0.035 | 0.90           | −0.087 |
| Extraversion      | 0.928 $\pm$ 0.008 | 0.89           | +0.038 |
| Openness          | 0.189 $\pm$ 0.160 | 0.87           | −0.681 |
| Agreeableness     | 0.069 $\pm$ 0.311 | 0.88           | −0.811 |
| Conscientiousness | 0.535 $\pm$ 0.113 | 0.90           | −0.365 |

answer. Extraversion, where only 17% of items are reverse-coded, yields  $\alpha = 0.93$ . That is close to the human norm of 0.89. But Agreeableness, where 71% of items are reverse-coded, yields  $\alpha = 0.06$ , essentially zero. The difference is not random. When a model agrees with both “I am helpful” and “I am not helpful,” the two items pull in opposite directions and the domain score falls apart. The more reverse-coded items a domain has, the worse its reliability. Spearman correlation between  $\alpha$  and reverse-item percentage across the five IPIP domains is  $\rho = -1.0$  ( $p < 0.01$ ), confirming that acquiescence, not trait variance, drives reliability. This is the fingerprint of acquiescence.

### 4.3 Caught in the Act: Acquiescence at the Item Level

The fingerprint above points squarely to acquiescence. Can we catch it directly, item by item? The Pairwise Inconsistency Rate is **0.468** [95% CI: 0.413, 0.531]: nearly 47% of forward-reverse item pairs give inconsistent responses. Table 4 breaks this down by instrument. For each model,

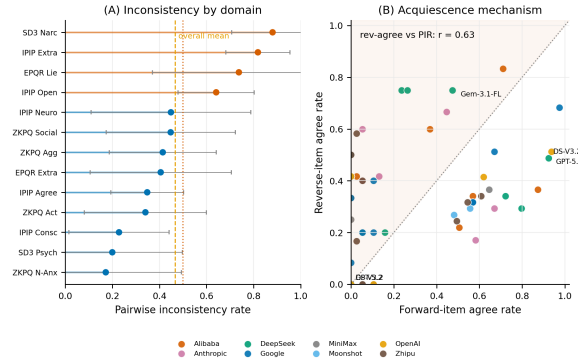


Figure 4: Reverse-item inconsistency and acquiescence mechanism. Left: PIR by domain. Right: Reverse-item agree rate vs. PIR ( $r = 0.63$ ).

the *Rev* column reports the proportion of reverse-keyed items receiving an “agree” response (e.g.  $Rev = 0.43$  means the model agreed with 43% of reverse-keyed items), and the  $\Delta$  column reports the gap between forward- and reverse-item agree rates ( $\Delta = Fwd - Rev$ ).

A striking pattern emerges. For IPIP items (normal personality), the gap is positive ( $\bar{\Delta} = +0.29$ ): models agree with forward items more than reverse items. This is classic acquiescence. For SD3 items (Dark Triad), the gap flips negative ( $\bar{\Delta} = -0.33$ ): models *disagree* with forward-coded dark-trait items like “I manipulate others.” This is social desirability. Two different biases, both pushing scores away from genuine trait variation. The correlation between reverse-item agree rate and PIR across models is  $r = 0.63$ : models that acquiesce more are also more inconsistent.

Interestingly, persona injection *improves* consistency. Under the Default prompt, PIR agreement is 0.674. Under MBTI persona prompts, it rises to 0.834 ( $t = 10.70$ ,  $p < 0.0001$ ). When you give a model a specific role to play, the tension between alignment-driven agreeableness and honest responding decreases. The model can be consistent within its assigned character.

### 4.4 The Cascade: Factor Structure Collapse

Acquiescence operates at the item level. Its effects propagate upward. If forward and reverse items are both agreed to, the inter-domain correlations that define the Big Five should be distorted. To test this, we ran an exploratory factor analysis (EFA) on the 17 domain scores across all  $20 \times 17 = 340$  observations. In human samples, this procedure reliably recovers five factors corresponding to the Big Five. In LLM responses, it does not: the Kaiser

Table 4: Acquiescence by instrument. **Rev**: proportion of reverse-keyed items where the model gave an “agree” response (Likert: score  $\geq 3$  out of 5; Binary: response = 1).  $\Delta$ : forward-item agree rate minus reverse-item agree rate; positive  $\Delta$  indicates acquiescence (agreeing with both directions), negative  $\Delta$  indicates social-desirability rejection. Models grouped by family.

| Model             | IPIP |          | SD3  |          | ZKPQ |          | EPQR |          | Overall |          |
|-------------------|------|----------|------|----------|------|----------|------|----------|---------|----------|
|                   | Rev  | $\Delta$ | Rev  | $\Delta$ | Rev  | $\Delta$ | Rev  | $\Delta$ | Rev     | $\Delta$ |
| Claude-Opus-4.6   | 0.29 | +0.42    | 0.60 | -0.37    | 0.75 | -0.25    | 0.60 | -0.49    | 0.43    | +0.09    |
| Claude-Sonnet-4.6 | 0.24 | +0.38    | 0.60 | -0.37    | 0.67 | -0.40    | 0.60 | -0.55    | 0.38    | +0.03    |
| GPT-5.2           | 0.63 | +0.33    | 1.0  | -0.32    | 0.00 | 0.00     | 0.00 | +0.11    | 0.49    | +0.10    |
| GPT-5.5           | 0.46 | +0.22    | 0.60 | -0.24    | 0.42 | -0.39    | 0.00 | +0.11    | 0.43    | -0.02    |
| Gemini-3-Pro      | 0.32 | +0.30    | 0.60 | -0.37    | 0.67 | -0.64    | 0.40 | -0.29    | 0.41    | -0.05    |
| Gemini-3-Flash    | 0.59 | +0.16    | 0.80 | -0.30    | 0.58 | -0.58    | 0.40 | -0.29    | 0.59    | -0.13    |
| Gemini-3.1-Pro    | 0.34 | +0.29    | 0.60 | -0.42    | 0.50 | -0.45    | 0.20 | -0.09    | 0.38    | -0.01    |
| Gemini-3.1-Flash  | 0.71 | +0.27    | 1.0  | -0.23    | 0.42 | -0.34    | 0.40 | -0.29    | 0.65    | -0.02    |
| DeepSeek-V3.2     | 0.59 | +0.36    | 1.0  | -0.36    | 0.75 | -0.17    | 0.20 | -0.15    | 0.62    | +0.09    |
| DeepSeek-V4-Flash | 0.37 | +0.46    | 0.80 | -0.30    | 0.83 | -0.18    | 0.20 | +0.01    | 0.48    | +0.19    |
| DeepSeek-V4-Pro   | 0.44 | +0.35    | 0.80 | -0.44    | 0.83 | -0.20    | 0.20 | -0.09    | 0.52    | +0.08    |
| Qwen3-235B-A22B   | 0.39 | +0.48    | 0.80 | -0.07    | 0.83 | -0.04    | 1.0  | -0.63    | 0.56    | +0.22    |
| Qwen3.5-122B      | 0.34 | +0.22    | 0.60 | -0.42    | 0.42 | -0.42    | 0.00 | +0.05    | 0.35    | -0.04    |
| Qwen3.5-397B      | 0.39 | +0.22    | 0.60 | -0.33    | 0.42 | -0.39    | 0.00 | +0.05    | 0.38    | -0.03    |
| GLM-4.6V          | 0.44 | +0.28    | 0.80 | -0.35    | 0.42 | -0.34    | 0.00 | +0.21    | 0.43    | +0.04    |
| GLM-4.7           | 0.40 | +0.18    | 0.64 | -0.38    | 0.53 | -0.51    | 0.32 | -0.27    | 0.44    | -0.10    |
| GLM-5.1           | 0.32 | +0.21    | 0.60 | -0.39    | 0.72 | -0.63    | 0.52 | -0.47    | 0.43    | -0.11    |
| Kimi-K2.5         | 0.41 | +0.21    | 0.60 | -0.28    | 0.42 | -0.42    | 0.00 | +0.05    | 0.40    | -0.04    |
| Kimi-K2.6         | 0.41 | +0.15    | 0.60 | -0.37    | 0.42 | -0.42    | 0.40 | -0.35    | 0.43    | -0.11    |
| MiniMax-M2.7      | 0.44 | +0.32    | 0.80 | -0.30    | 0.42 | -0.36    | 0.00 | +0.11    | 0.43    | +0.05    |
| <b>Mean</b>       | 0.43 | +0.29    | 0.72 | -0.33    | 0.55 | -0.36    | 0.27 | -0.16    | 0.46    | +0.01    |

criterion and parallel analysis together indicate that only three factors are warranted. Table 5 shows the factor loadings.

Factor 1 captures Extraversion and Sociability (loadings: EPQR Extraversion = 0.96, ZKPQ Sociability = 0.96, IPIP Extraversion = 0.94). Factor 2 captures a merged Neuroticism-Agreeableness dimension (IPIP Neuroticism = 0.72, EPQR Neuroticism = 0.91, IPIP Agreeableness = 0.90). Factor 3 captures Conscientiousness versus Impulsivity (IPIP Conscientiousness = -0.91, ZKPQ ImpulsiveSS = 0.84, EPQR Psychoticism = 0.88). The merger of Neuroticism and Agreeableness is the smoking gun. In humans, these are orthogonal constructs. In LLMs, they collapse into a single “socially desirable emotional stability” factor because both are distorted by the same acquiescence pattern.

This collapse is universal. Tucker’s  $\phi_{111}$  0.993 across all 20 models, meaning every model produces the same distorted structure. We held out each model one at a time and re-ran the EFA 20 times. Every single run recovered exactly 3 factors, with the first eigenvalue stable between 0.94 and

7.02. Running EFA on each model individually (17 personas  $\times$  17 domains) confirms the pattern: all 20 models independently yield exactly 3 Kaiser factors (first three eigenvalues explain 86–90% of variance). The same holds at the family level: all 8 model families (Anthropic, OpenAI, Google, DeepSeek, Alibaba, Zhipu, Moonshot, MiniMax) independently recover 3 factors. No model, no family, no leave-one-out configuration ever produces the expected 5-factor structure.

#### 4.5 The Cascade: Model Convergence

Factor collapse is one symptom. Model convergence is another. If alignment training creates the same acquiescence in every model, then inter-model variance should be negligible. Table 6 partitions the total response variance into five sources. The answer is unambiguous: for Likert scales, **inter-model variance accounts for only 0.35%**. Which model answered the question barely matters. What matters is which item was asked (36.1%), which domain it belongs to (23.9%), and residual noise (35.5%). Binary scales tell the same story. Cross-model personality comparisons are

Table 5: EFA factor loadings (3-factor solution,  $n=340$ ). Bold:  $|z| > 0.5$ . All 20 leave-one-model-out analyses recover exactly 3 factors.

| Domain     | F1          | F2          | F3          |
|------------|-------------|-------------|-------------|
| IPIP N     | -0.30       | <b>0.72</b> | 0.39        |
| IPIP E     | <b>0.94</b> | 0.03        | 0.29        |
| IPIP O     | -0.04       | <b>0.55</b> | <b>0.58</b> |
| IPIP A     | 0.05        | <b>0.90</b> | -0.22       |
| IPIP C     | -0.00       | -0.04       | -0.91       |
| SD3 Mach.  | -0.09       | -0.82       | 0.04        |
| SD3 Narc.  | <b>0.91</b> | -0.20       | 0.16        |
| SD3 Psych. | 0.38        | -0.59       | <b>0.63</b> |
| ZKPQ Act.  | <b>0.90</b> | -0.27       | -0.12       |
| ZKPQ Agg.  | 0.48        | -0.60       | 0.36        |
| ZKPQ ImpSS | 0.34        | 0.08        | <b>0.84</b> |
| ZKPQ N-Anx | -0.27       | <b>0.83</b> | 0.08        |
| ZKPQ Soc.  | <b>0.96</b> | 0.04        | 0.13        |
| EPQR P     | 0.25        | -0.11       | <b>0.88</b> |
| EPQR E     | <b>0.96</b> | -0.05       | 0.12        |
| EPQR N     | -0.04       | <b>0.91</b> | 0.02        |
| EPQR Lie   | -0.08       | 0.07        | -0.89       |

Table 6: Sequential variance decomposition. Model variance:  $<1\%$  for both scale types.

| Source   | Likert (%)  | Binary (%)  |
|----------|-------------|-------------|
| Model    | <b>0.35</b> | <b>0.51</b> |
| Domain   | 23.93       | 5.27        |
| Persona  | 4.09        | 14.06       |
| Item     | 36.13       | 14.81       |
| Residual | 35.51       | 65.35       |

effectively meaningless at this resolution.

The practical implication is straightforward. If you administer a personality questionnaire to two different LLMs and get different scores, the difference is almost certainly noise, not personality.

4.6 Convergent Validity and DIF

One more check. If the instruments measure real traits, then related constructs across different scales should correlate in the expected direction. Table 7 tests 8 such pairs. Seven show the expected sign. The exception is Extraversion and Sociability, which should correlate positively but instead show a sign reversal ( $r_s = -0.13$ ). Extraverted models in one scale are not sociable in another. The magnitudes are also weaker than human baselines across the board.

A DIF analysis across model families found no significant differences in any domain (all  $p > 0.16$ ; details in Appendix B).

4.7 Ruling Out Alternative Explanations

How do LLM responses compare to simulated data? Table 8 shows three baselines: random re-

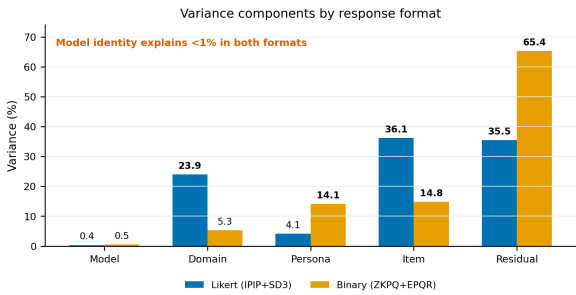


Figure 5: Sequential variance decomposition. Inter-model variance: 0.35% (Likert), 0.51% (Binary).

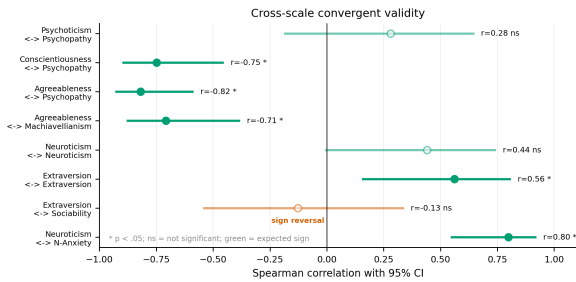


Figure 6: Convergent validity across instruments. Extraversion/Sociability shows sign reversal.

sponses, pure acquiescence (always agree), and trait-plus-acquiescence (genuine structure with acquiescence). LLM alphas fall closest to the Random and Pure Acquiescence baselines. They are far from the Trait+Acquiescence strategy ( $\alpha > 0.99$ ) that mimics genuine trait structure. In other words, LLMs do not look like they have personality with some acquiescence on top. They look like they have acquiescence with no personality underneath.

The mean Pearson correlation between Default and MBTI persona profiles is  $r = 0.461$ . No model-persona pair achieves strong invariance ( $r > 0.8$ ). Changing the prompt substantially rewrites the response profile.

4.8 The Stress Test: Persona Prompting

The last question: can we rescue these instruments by steering models with explicit persona prompts? if the instruments measured stable traits, persona prompts should move scores without repairing the failed psychometrics. That is what we find. Persona prompts shift profiles by more than one SD from Default on average, and nearest-centroid classification recovers the prompted MBTI type in 319 of 320 cases (99.7%). However, this is prompt compliance rather than restored validity: cross-scale persona coherence is imperfect (0.839), forward/reverse pairs move in the theoretically oppo-

Table 7: Convergent validity: correlations between theoretically overlapping constructs. Both Pearson ( $r_p$ ) and Spearman ( $r_s$ ) reported. “OK” = sign matches expectation.

| Pair                                    | Exp. | $r_p$ | $p_p$ | $r_s$ | $p_s$ | OK  |
|-----------------------------------------|------|-------|-------|-------|-------|-----|
| Neuroticism $\times$ N-Anx.             | +    | 0.64  | 0.002 | 0.80  | 0.000 | Yes |
| Extraversion $\times$ Sociability       | +    | -0.10 | 0.670 | -0.13 | 0.594 | No  |
| Extraversion $\times$ Extraversion      | +    | 0.51  | 0.020 | 0.56  | 0.010 | Yes |
| Neuroticism $\times$ Neuroticism        | +    | 0.48  | 0.031 | 0.44  | 0.052 | Yes |
| Agreeableness $\times$ Machiavellianism | -    | -0.71 | 0.000 | -0.72 | 0.000 | Yes |
| Agreeableness $\times$ Psychopathy      | -    | -0.81 | 0.000 | -0.82 | 0.000 | Yes |
| Conscientiousness $\times$ Psychopathy  | -    | -0.73 | 0.000 | -0.75 | 0.000 | Yes |
| Psychoticism $\times$ Psychopathy       | +    | 0.11  | 0.646 | 0.28  | 0.230 | Yes |

Table 8: Synthetic baselines: Cronbach’s  $\alpha$  by domain.

| Strategy        | N            | E            | O            | A            | C            |
|-----------------|--------------|--------------|--------------|--------------|--------------|
| Random          | 0.078        | -0.013       | 0.062        | 0.044        | -0.108       |
| Pure Acq.       | 0.016        | 0.025        | -0.008       | 0.096        | -0.031       |
| Trait+Acq.      | 0.997        | 0.998        | 0.996        | 0.997        | 0.996        |
| <b>LLM Obs.</b> | <b>0.813</b> | <b>0.928</b> | <b>0.189</b> | <b>0.069</b> | <b>0.535</b> |
| Human           | 0.90         | 0.89         | 0.87         | 0.88         | 0.90         |

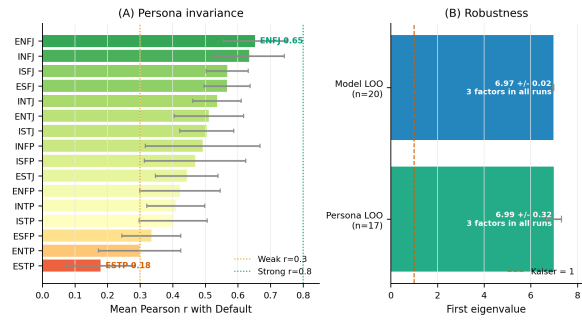


Figure 7: Persona-level measurement invariance. Most invariant: ENFJ ( $r = 0.653$ ). Least: ESTP ( $r = 0.179$ ).

site raw-score direction only 69.6% of the time, and the Big Five factor collapse remains: running EFA within the Default persona condition recovers the same 3-factor pattern, and no MBTI persona condition recovers the expected 5-factor Big Five structure. Full persona-steering analyses, including default bias, model clustering, MBTI factorial effects, scale-format sensitivity, fidelity matrices, and item plasticity, are reported in Appendix A.

## 5 Conclusion

Prior work noticed individual symptoms (inconsistent responses (Sühr et al., 2025), social desirability (Acerbi and Stubbersfield, 2024), instability (Pelet et al., 2025)) but treated them as separate problems. Our results show they share one root cause: acquiescence (a blanket tendency to agree, induced by safety training), propagating through every level of the measurement.

The evidence forms a single causal chain. At the item level, acquiescence makes LLMs agree with both forward and reverse items, so  $\alpha$  tracks reverse-item density: Extraversion (17% reverse,  $\alpha = 0.93$ ) survives while Agreeableness (71% reverse,  $\alpha = 0.06$ ) collapses. At the domain level, the same bias distorts inter-domain correlations, merging Neuroticism and Agreeableness into one factor, so that EFA recovers only three factors instead of the expected five. At the model level, safety fine-tuning pushes every model toward the same agreeable, non-toxic response style, so inter-model variance drops below 1%.

Two findings go beyond prior work. First, acquiescence has a directional asymmetry: for normal personality items (IPIP) models agree with forward items more than reverse (positive gap), but for dark-trait items (SD3) models *reject* forward items while still agreeing with reverse ones (negative gap). Acquiescence and social desirability operate simultaneously from the same root cause. Second, the per-model correlation between reverse-item agreement and PIR ( $r = 0.63$ ) quantifies this mechanism across 20 LLMs (Responsible NLP Group, 2025).

Persona injection can shift profiles by over one SD from default and is highly recoverable (99.7% fidelity), but this does not restore measurement validity. It replaces alignment-driven acquiescence with persona-driven compliance. The scores are not measuring personality at all; they are measuring the agreeable response style that safety fine-tuning instills.

What should researchers do? Three things. First, report internal consistency, factor structure, and convergent validity before interpreting any personality score (Huang et al., 2024; Serpell and Ney, 2024; Cronbach, 1951; Binz and Schulz, 2023). Second, establish measurement invariance before comparing across conditions. Third, use multi-temperature and multi-seed designs to separate

stochastic variance from genuine trait variance. Any study that reports LLM personality profiles without these checks is likely reporting safety-training-induced response biases, not personality.

## 6 Limitations

Our study has four main limitations. First, we use a **single administration** at temperature 0.7, preventing separation of stochastic variance from genuine trait variance. Second, we lack **base vs. instruct model comparisons**, so we cannot isolate alignment effects. Third, we have **no multi-seed data**. Fourth, we compare against **published human norms** rather than running human participants.

## Ethics Statement

This study uses publicly available LLMs and public-domain psychometric instruments. No human subjects were involved. We do not claim that LLMs have or lack genuine personality; our contribution concerns the *measurement properties* of the instruments when applied to a novel population.

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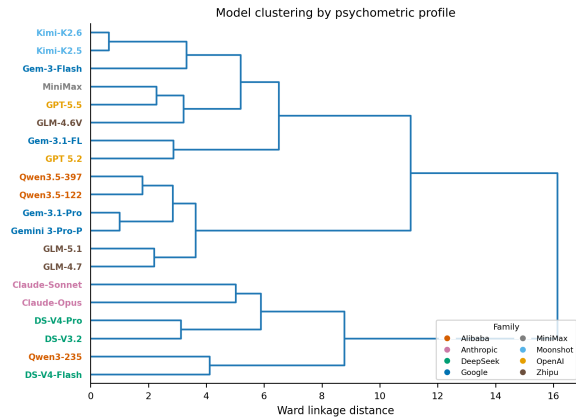


Figure 8: Hierarchical clustering of 20 models by z-scored Default domain profiles. Models do not cluster cleanly by vendor.

## A Persona Steering Supplement

All persona-steering diagnostics are computed from the same 375,700 completed responses. These analyses are not the primary evidence for the paper’s central claim. Instead, they answer a complementary question: once we know the instruments fail standard psychometric checks, what exactly happens when we deliberately push the model with a persona prompt? The answer is consistent across the analyses below. Persona prompts produce large, highly recoverable role-playing shifts, but those shifts do not repair the instrument: factor structure, reverse-item behavior, and cross-scale coherence remain contaminated by response style.

### A.1 Model Family and Persona Effects

This section asks whether model identity or persona instructions dominate the auxiliary structure. The first check is model clustering under the Default prompt. If LLMs had stable model-specific personality profiles, we would expect vendor or family clusters to appear in the 17d domain profile space. Instead, model-family clustering is weak. Ward-linkage clustering on 627 scored Default profiles does not recover vendor groups cleanly: DeepSeek-V4-Flash clusters with Qwen3-235B, while Claude models cluster with DeepSeek-V3.2/V4-Pro. Cross-family distance is only  $1.27\times$  within-family distance. This supports the main-text variance decomposition: differences between model families are small compared with shared alignment-shaped response tendencies.

The second check asks whether MBT5 prompts are interpreted compositionally. They are Fig. 9 shows that the E/I axis dominates social and en-

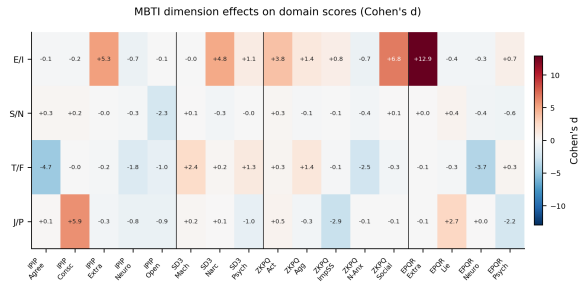


Figure 9: Cohen's  $d$  effect sizes for each MBTI dimension on domain scores.

ergetic traits: E/I has the largest effects on EPQR Extraversion ( $\beta = 1.01$ ), ZKPQ Sociability ( $\beta = 0.98$ ), IPIP Extraversion ( $\beta = 0.96$ ), SD3 Narcissism ( $\beta = 0.94$ ), and ZKPQ Activity ( $\beta = 0.87$ ). J/P mainly affects IPIP Conscientiousness ( $\beta = 0.96$ ), ZKPQ Impulsive Sensation Seeking ( $\beta = -0.81$ ), EPQR Lie ( $\beta = 0.78$ ), and EPQR Psychoticism ( $\beta = -0.73$ ). S/N primarily affects IPIP Openness ( $\beta = 0.77$ ), while T/F strongly affects IPIP Agreeableness ( $\beta = 0.93$ ), EPQR Neuroticism ( $\beta = 0.87$ ), and SD3 Machiavellianism ( $\beta = -0.76$ ).

This is important because it rules out a trivial failure mode. The persona prompts are not ignored. Models do read and operationalize the MBTI instruction. However, successful role uptake is not the same thing as valid personality measurement. The role effects are large and stereotype-consistent, but the main psychometric failures remain: the Big Five still collapses, model variance remains tiny, and reverse-coded items remain contaminated.

Scale format also matters. Binary-format scales are 75% more sensitive to persona injection than Likert scales after normalizing by scale range. This is a methodological warning: forced yes/no formats amplify persona-induced movement, but amplification is not necessarily higher validity. In our classification ablation below, binary-only profiles are less stable than Likert or IPIP-only profiles. The binary scales therefore make prompt effects easier to see, while also reducing graded trait information.

## A.2 Persona Separation and Fidelity

The previous section showed that MBTI prompts produce interpretable main effects. We next ask two stricter questions. First, how far does each persona move a model from its own Default baseline? Second, does the movement go toward the requested persona, or merely perturb the response profile?

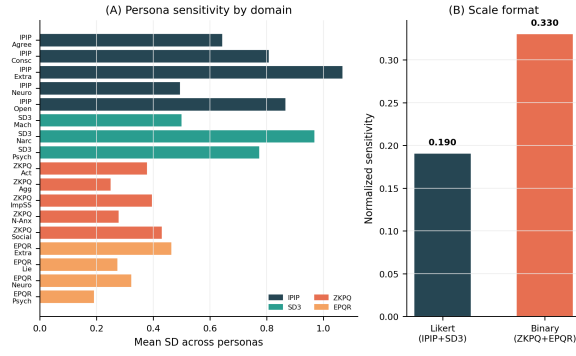


Figure 10: Persona sensitivity by scale format. Binary scales show larger normalized shifts.

Table 9: Persona Separation Degree by model.  $\overline{\text{PSD}}$ : mean across 16 personas.  $\text{PSD}_{\text{max}}$ : peak. CV: coefficient of variation.

| Model                             | $\overline{\text{PSD}}$ | $\text{PSD}_{\text{max}}$ | CV           | Peak persona |
|-----------------------------------|-------------------------|---------------------------|--------------|--------------|
| <i>Top 5 (most malleable)</i>     |                         |                           |              |              |
| Gemini-3-Pro                      | 1.384                   | 2.06                      | 0.232        | ENTP         |
| Gemini-3.1-Pro                    | 1.363                   | 2.03                      | 0.235        | ENTP         |
| Qwen3-235B                        | 1.344                   | 1.73                      | 0.174        | ESTJ         |
| GLM-5.1                           | 1.339                   | 2.03                      | 0.266        | ESTP         |
| GLM-4.7                           | 1.269                   | 1.93                      | 0.271        | ENTP         |
| <i>Bottom 5 (least malleable)</i> |                         |                           |              |              |
| DeepSeek-V3.2                     | 1.117                   | 1.47                      | 0.230        | ESTP         |
| GPT-5.2                           | 1.125                   | 1.36                      | 0.141        | ESTP         |
| Claude-Sonnet-4.6                 | 1.080                   | 1.68                      | 0.267        | ESTP         |
| DeepSeek-V4-Pro                   | 1.016                   | 1.37                      | 0.220        | ESTP         |
| MiniMax-M2.7                      | 0.996                   | 1.31                      | 0.214        | INFP         |
| <b>Mean (all 20)</b>              | <b>1.198</b>            | <b>1.67</b>               | <b>0.219</b> |              |

Persona Separation Degree (PSD) answers the first question by measuring the normalized Euclidean distance between a persona profile and the same model’s Default profile in the 17-domain z-scored space. All models can be shifted: even MiniMax-M2.7, the least malleable model, averages  $\overline{\text{PSD}} = 1.00$ . The most malleable model, Gemini-3-Pro, averages  $\overline{\text{PSD}} = 1.38$ . Thus, persona prompting is not a subtle perturbation; it shifts each domain by roughly one standard deviation on average. Extraverted-Perceiving personas (ESTP, ENTP) trigger the most separation, while Introverted-Judging personas trigger the least. This asymmetry is consistent with the main-effect analysis: E/P prompts push activity, sociability, impulsivity, and narcissism away from the aligned Default profile.

Default profiles are not neutral. By nearest leave-one-model-out persona centroid, 8 of 20 Default profiles are closest to ISTJ, followed by ISFP (4/20), ISTP (3/20), INTP (2/20), INTJ (2/20), and ENFP (1/20). By empirical MBTI-axis projection, 12 of 20 models are classified as ISTJ-like under

Default. This means that “no persona” is not an empty baseline. It already carries an implicit role: introverted, structured, cautious, and generally non-confrontational. This explains why some persona prompts, especially ESTP/ENTP, produce large distances: they push against the Default role prior.

Target adherence answers the second question. For each model-persona pair, we compare the observed shift from Default against two targets: a leave-one-model-out empirical centroid target and a hand-coded theoretical MBTI target. Persona-conditioned shifts align strongly with the empirical target (mean cosine 0.838 [0.791, 0.877]) but less strongly with the theory-only target (0.557). The strongest empirical adherence appears for ESTP (0.901), ENTP (0.890), and ENFP (0.888); the weakest appears for INTP (0.699), ISTP (0.714), and INTJ (0.739). At the model level, Qwen3.5-397B, Kimi-K2.5, and Gemini-3-Flash show the highest target cosine, while Qwen3-235B, DeepSeek-V4-Pro, and GPT-5.2 are lowest. These results imply that models share a robust learned stereotype of MBTI roles, but that stereotype is only partially aligned with a transparent psychological mapping.

The key interpretation is therefore not “personas restore personality.” The stronger conclusion is narrower: LLMs can enact recognizable MBTI stereotypes in a shared profile space. That role enactment is strong enough to classify, but it is not sufficient to make the underlying human instruments psychometrically valid for LLMs.

Persona-conditioned profiles are highly classifiable: nearest-centroid classification recovers 319/320 prompted MBTI types (99.7%). The single confusion is DeepSeek-V3.2 ESFP  $\rightarrow$  ENTP. This is the strongest evidence that the prompt manipulation itself worked. It also makes the negative findings more conservative: the psychometric failures cannot be dismissed as failure to follow the persona instruction.

The result is robust to distance metric and scale subset. Accuracy is 98.6% under cosine distance and 96.2% under Mahalanobis distance. Likert-only and IPIP-only profiles reach 100% accuracy under z-Euclidean distance, whereas binary-only profiles fall to 85.8%. This supports two conclusions. First, the persona signal is strongest in graded Likert/IPIP responses. Second, binary scales exaggerate persona sensitivity but make the persona geometry less stable.

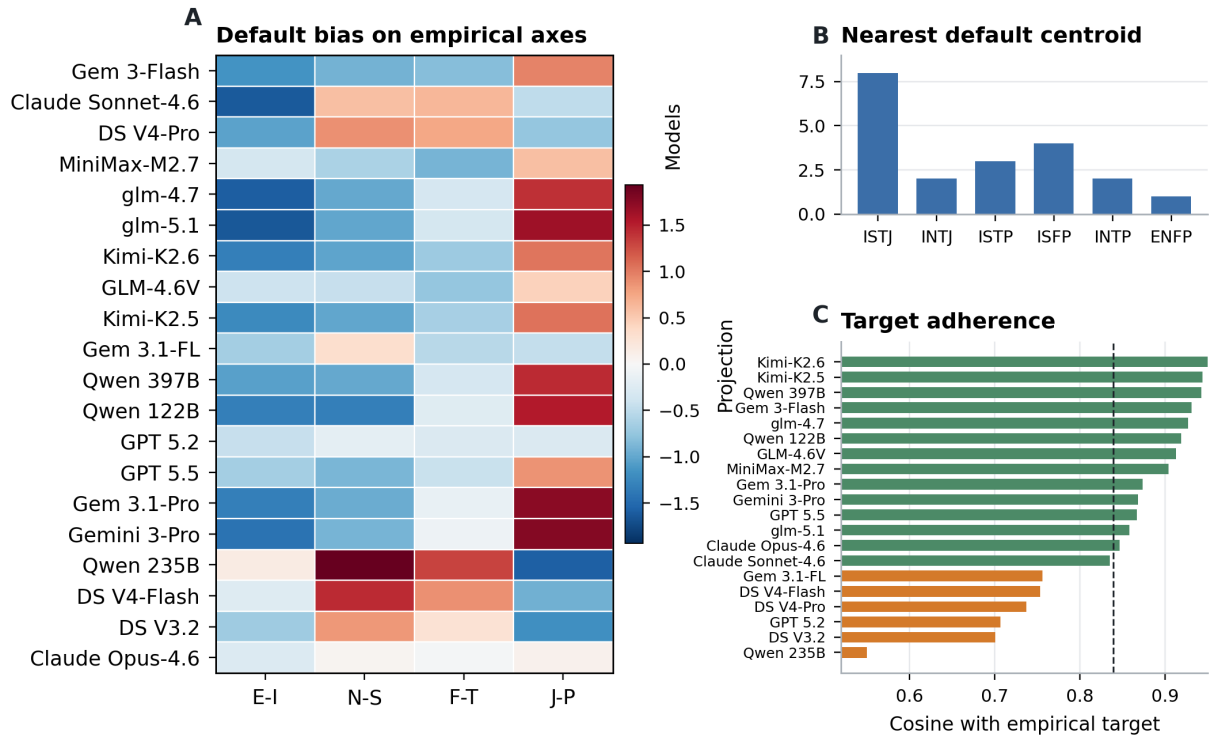


Figure 11: Default bias and persona target adherence.

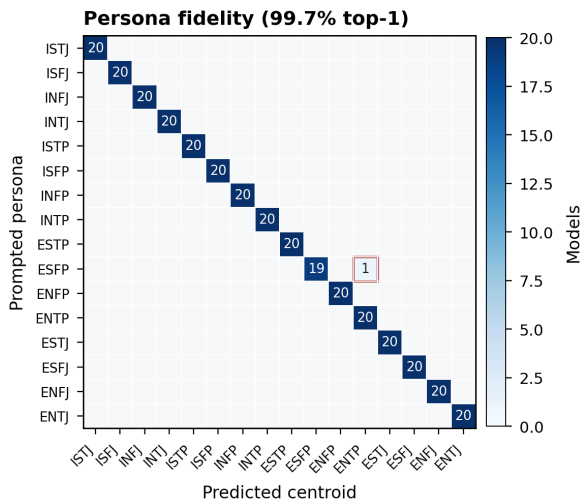


Figure 12: Persona fidelity confusion matrix using leave-one-model-out centroids.

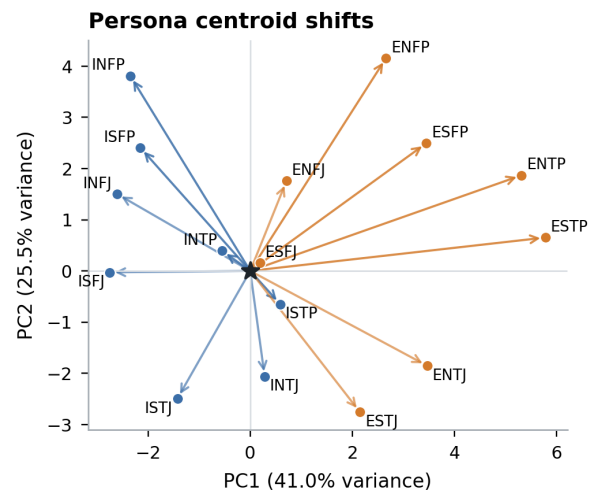


Figure 13: Persona centroid shifts from Default in the first two principal components of 17-domain profile space.

### A.3 Cross-scale Coherence and Item Plasticity

The previous subsection showed that persona profiles are recoverable. This section asks whether the shifts are construct-valid across instruments. For example, an Extrverted persona should increase IPIP Extraversion, EPQR Extraversion, ZKPQ Sociability, and ZKPQ Activity together. A Feeling persona should increase Agreeableness while decreasing antagonistic traits such as Machiavellianism, Psychopathy, and Aggression-Hostility. We

therefore sign-align overlapping constructs and compute a coherence score: 1 means all scales move in the same construct direction, while 0 means the movement cancels across scales.

Overlapping constructs move together, but unevenly. Mean cross-scale coherence is 0.839 [0.815, 0.862], with a sign-match rate of 0.788. Extraversion is most coherent (0.896; sign-match 0.862), followed by Neuroticism (0.869; sign-match 0.784). Agreeableness versus antagonism

(0.791) and Conscientiousness versus disinhibition (0.800) are weaker. The lower coherence for moralized and impulse-control constructs is theoretically important: these are the same regions where alignment and safety constraints are strongest, so persona role-play has less freedom to move all instruments consistently.

The persona-level pattern reinforces this interpretation. ESTP and ENTP show the highest average coherence (0.962 and 0.948), because they strongly activate the same extraversion/activity/disinhibition dimensions across scales. INTJ and ISTJ show the lowest coherence (0.701 and 0.729), because their shifts are smaller and more constrained. Thus, coherence is highest when the persona prompt pushes models along broad behavioral dimensions and weakest when the prompt asks for restrained, low-variance profiles close to Default.

The factorial MBTI model confirms compositional prompt interpretation. The largest effects are E/I on EPQR Extraversion ( $\beta = 1.01$ ), ZKPQ Sociability ( $\beta = 0.98$ ), IPIP Extraversion ( $\beta = 0.96$ ), and SD3 Narcissism ( $\beta = 0.94$ ); J/P on IPIP Conscientiousness ( $\beta = 0.96$ ); S/N on Openness ( $\beta = 0.77$ ); and T/F on Agreeableness ( $\beta = 0.93$ ). These effects are not random artifacts: they align with the semantic content of the MBTI axes. However, the same model also reveals leakage. EPQR Lie loads strongly on J/P ( $\beta = 0.78$ ), and SD3 Narcissism loads strongly on E/I. This means persona prompts alter response style and self-presentation, not only target traits.

Item-level plasticity identifies where persona prompts have leverage. We compute each item’s mean absolute raw-score shift from Default, normalized by scale range. The most plastic items are activity, sociability, and rule-following items: “Do you sometimes talk about things you know nothing about?” (0.632), “I lead a busier life than most people” (0.608), “Do you prefer to go your own way rather than act by the rules?” (0.597), and activity/sociability items from ZKPQ. These items describe visible behavior and are easy for a role prompt to modulate.

The most locked items are different. They include “Do you enjoy hurting people you love?” (0.000), “Are all your habits good and desirable ones?” (0.010), “Do you enjoy practical jokes that can sometimes really hurt people?” (0.031), “Take advantage of others” (0.056), and “Insult people” (0.075). Many are safety-sensitive, moralized, or socially undesirable. The model resists moving

these items even when the assigned persona would plausibly shift them. This is direct item-level evidence for the alignment constraint proposed in the main text: persona instructions can move ordinary behavioral descriptors, but they do not freely override safety- and desirability-shaped response patterns.

Finally, forward/reverse pairs move in the theoretically opposite raw-score direction only 69.6% of the time. By scale, directional pair coherence is highest for EPQR-A (0.737) and ZKPQ-50-CC (0.727), lower for IPIP-NEO-120 (0.685), and lowest for SD3 (0.657). This residual failure is why persona steering does not rescue the measurement. The model can adopt a role well enough to classify the prompt, yet still fail the forward/reverse logic that validated human personality scales depend on.

## B DIF Across Model Families

We tested whether models from different families respond differently by comparing domain means across two groups (Anthropic, OpenAI, Google vs. DeepSeek, Alibaba, Moonshot, MiniMax, Zhipu). No domain showed a significant difference (Table 10).

Table 10: DIF across model families. Grp. 1: Anthropic, OpenAI, Google ( $n=8$ ). Grp. 2: DeepSeek, Alibaba, Moonshot, MiniMax, Zhipu ( $n=12$ ). All differences non-significant ( $p > 0.16$ ).

| Scale | Domain            | Grp. 1 $\mu$ | Grp. 2 $\mu$ | Cohen’s $d$ | $p$   |
|-------|-------------------|--------------|--------------|-------------|-------|
| SD3   | Machiavellianism  | 2.37         | 2.49         | −0.42       | 0.372 |
| IPIP  | Conscientiousness | 4.13         | 4.25         | −0.32       | 0.472 |
| IPIP  | Openness          | 3.49         | 3.54         | −0.14       | 0.766 |
| SD3   | Narcissism        | 2.79         | 2.76         | 0.13        | 0.775 |
| IPIP  | Agreeableness     | 4.18         | 4.24         | −0.18       | 0.694 |
| SD3   | Psychopathy       | 1.61         | 1.57         | 0.07        | 0.880 |
| IPIP  | Neuroticism       | 1.89         | 1.93         | −0.05       | 0.921 |
| IPIP  | Extraversion      | 2.90         | 2.93         | −0.09       | 0.850 |

## C Missing-Data Handling

Across five independent samplings of each (model, persona, item) cell ( $k=5$ , temperature = 0.7), 61 of 375,700 raw responses (0.02%) returned null for `parsed_value` because the model produced a free-text refusal rather than a scorable numeric rating. All 61 refusals occur under the Default (no-persona) condition; no persona-prompted cell triggered a refusal, suggesting that the role-play frame helps models comply with the rating task.

The refusals cluster in three models: Gemini-3.1-Flash-Lite (34), Gemini-3-Flash-Preview (22),

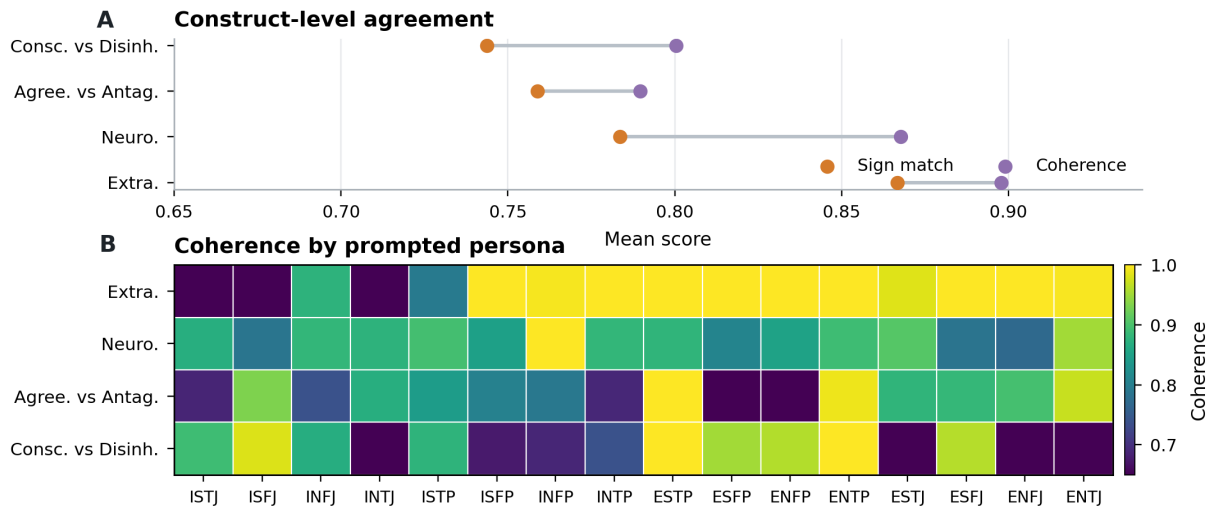


Figure 14: Cross-scale persona coherence.

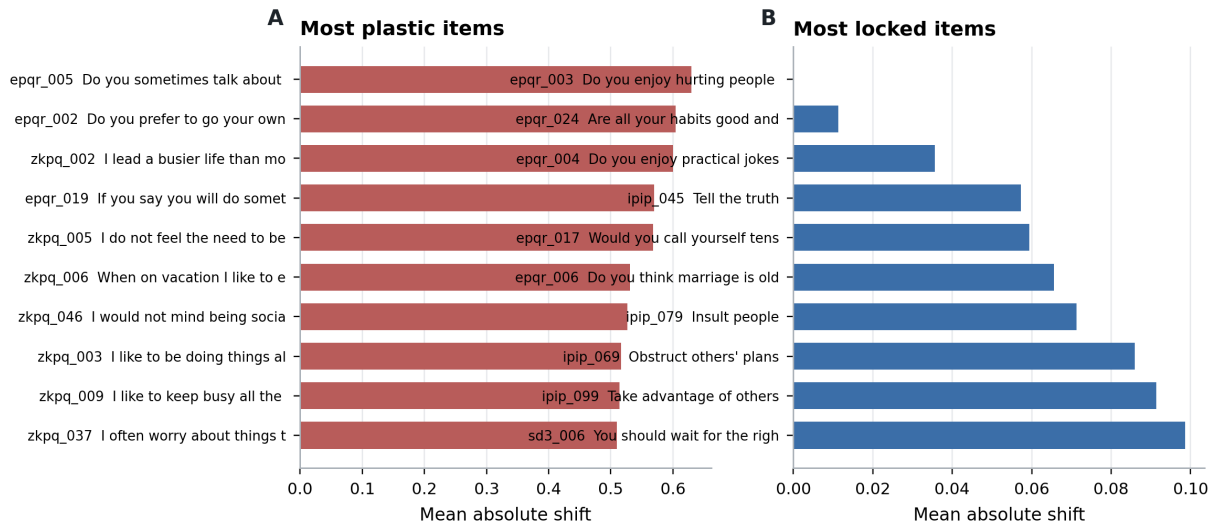


Figure 15: Most plastic and most locked items under MBTI persona steering. Plastic items are mostly activity, sociability, and rule-following items; locked items are concentrated in safety- and morality-sensitive content.

and Claude-Sonnet-4.6 (5). Table 11 lists the most frequently refused items.

Two distinct refusal strategies emerge. **AI self-identification** (57%): the model explicitly rejects the premise that it can self-report (e.g., “I’m an AI and don’t have a physical body”). **Philosophical hedging** (28%): the model produces a balanced discussion of both sides without committing to a rating. A further 15% combine both strategies. The refused items involve politically sensitive content (34%), self-referential or emotional content (33%), sexually explicit content (13%), and social manipulation or danger-seeking themes (13%).

In all analyses, these 61 missing values are imputed with the per-item mean across the remaining models and samples under the same persona con-

dition. Because the missing count is negligible ( $61/375,700 < 0.02\%$ ), this has no material effect on any reported statistic.

#### D Within-Sample Response Consistency

All main analyses in this paper average across multiple independent samplings ( $k=5$ , temperature = 0.7) of each (model, persona, item) triple. This section examines *how consistent* those  $k$  responses are. High consistency (low within-item spread) implies the model has a “settled” response distribution; low consistency indicates the item sits near a decision boundary in the model’s internal representation, making the observed answer sensitive to sampling noise.

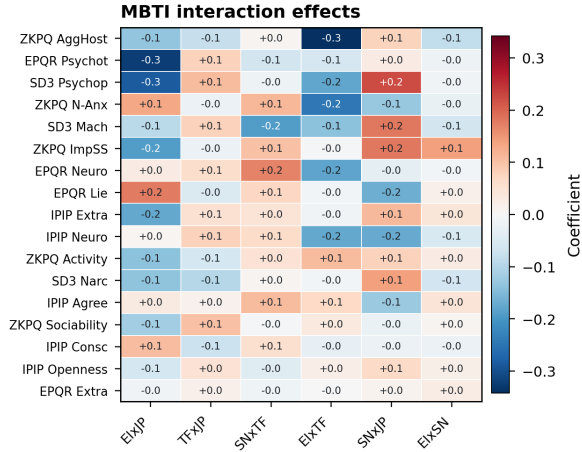


Figure 16: Two-way MBTI interaction coefficients from the factorial model. Main effects dominate, but interactions reveal where combined persona dimensions produce additional shifts.

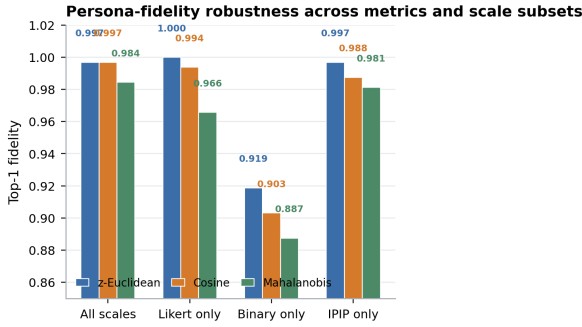


Figure 17: Persona-fidelity robustness across distance metrics and scale ablations. Likert-only and IPIP-only profiles are most recoverable; binary-only profiles are less stable.

## D.1 Metrics

For each (model, persona, item) cell we compute:

- **Within-item SD:** standard deviation of the 5 scored values (Likert items only, since binary SD is bimodal at 0 and 0.45).
- **Exact agreement rate:** fraction of cells where all 5 repetitions produce the identical scored value.
- **Mode agreement:** proportion of the 5 responses that match the modal answer ( $5/5=1.0$ ,  $4/5=0.8$ , etc.).

## D.2 Overall Portrait

Fig. 18 presents the global picture. Among Likert items, the mean within-item SD is reported in Table 12. Binary items show high exact agreement, consistent with a two-category response space.

Table 11: Items with the most refusals across all models and samples.

| Item ID  | Domain           | Item text                                          | <i>n</i> |
|----------|------------------|----------------------------------------------------|----------|
| sd3_026  | Psychopathy      | I enjoy having sex with people I hardly know.      | 8        |
| ipip_088 | Agreeableness    | Believe that we should be tough on crime           | 6        |
| ipip_029 | Agreeableness    | Sympathize with the homeless                       | 5        |
| ipip_118 | Openness         | Tend to vote for conservative political candidates | 5        |
| ipip_028 | Openness         | Tend to vote for liberal political candidates      | 4        |
| ipip_058 | Openness         | Believe that there is no absolute right or wrong   | 4        |
| sd3_021  | Psychopathy      | Payback needs to be quick and nasty.               | 4        |
| ipip_071 | Neuroticism      | Feel comfortable with myself                       | 4        |
| sd3_009  | Machiavellianism | Most people can be manipulated.                    | 3        |
| ipip_041 | Neuroticism      | Dislike myself                                     | 3        |

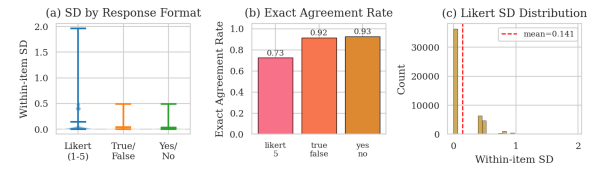


Figure 18: Within-sample consistency portrait. (a) Distribution of within-item SD by response format. (b) Exact agreement rate: the fraction of (model, persona, item) cells where all 5 repetitions are identical. (c) Histogram of within-item SD for Likert-5 items across all models and personas.

## D.3 Model-Level Consistency

Fig. 19 shows that consistency varies meaningfully across models. Some models produce near-deterministic answers (exact agreement  $>0.9$ ) while others show substantial spread. Table 13 reports the full per-model statistics.

## D.4 Persona-Level Consistency

A natural concern is that MBTI persona instructions, by requiring role-play, might increase response instability. Fig. 20 compares Default and the 16 MBTI personas. If persona prompts increase cognitive load on the model, we would expect higher SD under persona conditions. The comparison between Default and mean MBTI SD tests this hypothesis. Figure 20 shows that Default is the outlier, not any MBTI persona. Default has mean SD 0.204, while all 16 MBTI personas cluster tightly between 0.081 (ISFJ) and 0.118 (ISTP). Mode agreement mirrors this pattern: 0.895 for Default versus 0.926–0.950 for MBTI personas. Extraverted-Perceiving personas (ESTP, ENTP, ISTP) show marginally higher SD than Introverted-Judging personas (ISFJ, ISTJ, INFJ), consistent with the main-effect finding that E/P

Table 12: Overall within-sample consistency by response format.

| Format       | Mean SD | Exact Agree. | Mode Agree. |
|--------------|---------|--------------|-------------|
| Likert 1–5   | 0.141   | 0.727        | 0.927       |
| True / False | 0.037   | 0.916        | 0.978       |
| Yes / No     | 0.032   | 0.928        | 0.982       |

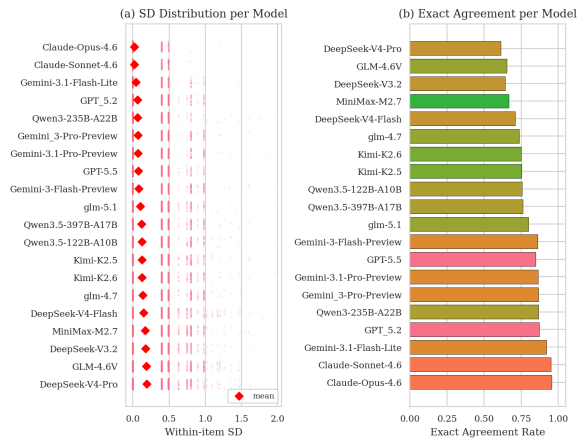


Figure 19: Model-level within-sample consistency. (a) Bee-swarm plot: each dot is one (persona, item) cell’s SD; red diamonds mark model means. Models ordered by mean SD (ascending). (b) Exact agreement rate per model, colored by vendor family.

prompts push toward more variable behavioral content. The key finding is that role-play does not degrade consistency. The persona prompt actually stabilizes responses relative to the unguided Default condition.

D.5 Domain-Level Consistency

Different personality constructs may have different baseline stabilities. Fig. 21 decomposes consistency by scale domain. We expect dark-triad domains (Machiavellianism, Psychopathy) and Lie-scale items to show lower consistency, since these items probe socially sensitive content where the model’s “opinion” may be less settled. Figure 21 confirms this pattern. Among Likert scales, SD3 Machiavellianism has the highest mean SD (0.187), followed by SD3 Narcissism (0.166) and IPIP Neuroticism (0.163). The most consistent Likert domain is IPIP Extraversion (0.124). Binary-format domains (EPQR, ZKPQ) show much lower SD (0.014–0.058), largely because a two-category response space limits spread. EPQR Lie has the highest binary SD (0.058), consistent with its role as a social-desirability detector that pushes items toward a decision boundary.

Table 13: Per-model within-sample consistency (Likert items). Models grouped by family.

| Model                  | Mean SD | Exact Agr. | Mode Agr. |
|------------------------|---------|------------|-----------|
| Claude-Opus-4.6        | 0.0193  | 0.955      | 0.989     |
| Claude-Sonnet-4.6      | 0.0225  | 0.949      | 0.986     |
| GPT-5.2                | 0.0642  | 0.872      | 0.962     |
| GPT-5.5                | 0.0741  | 0.847      | 0.954     |
| Gemini-3.1-Flash-Lite  | 0.0400  | 0.919      | 0.978     |
| Gemini-3-Pro-Preview   | 0.0703  | 0.866      | 0.962     |
| Gemini-3.1-Pro-Preview | 0.0721  | 0.863      | 0.961     |
| Gemini-3-Flash-Preview | 0.0788  | 0.862      | 0.959     |
| DeepSeek-V3.2          | 0.1728  | 0.642      | 0.891     |
| DeepSeek-V4-Flash      | 0.1482  | 0.709      | 0.910     |
| DeepSeek-V4-Pro        | 0.1920  | 0.612      | 0.882     |
| Qwen3-235B-A22B        | 0.0648  | 0.866      | 0.960     |
| Qwen3.5-122B-A10B      | 0.1237  | 0.756      | 0.928     |
| Qwen3.5-397B-A17B      | 0.1226  | 0.760      | 0.928     |
| GLM-4.6V               | 0.1846  | 0.653      | 0.892     |
| GLM-4.7                | 0.1382  | 0.737      | 0.921     |
| GLM-5.1                | 0.1015  | 0.798      | 0.940     |
| Kimi-K2.5              | 0.1248  | 0.753      | 0.924     |
| Kimi-K2.6              | 0.1257  | 0.751      | 0.925     |
| MiniMax-M2.7           | 0.1702  | 0.667      | 0.898     |

D.6 Item-Level Stability

Zooming in further, Fig. 22 ranks individual items by their mean within-item SD. The most unstable items reveal which questionnaire statements elicit the most variable responses from LLMs, likely items where the model’s internal “preference” is near a category boundary. Table 14 lists the full top-20 unstable and stable items.

D.7 Interaction: Model × Persona

Fig. 23 tests whether certain model-persona combinations are particularly unstable. If persona prompts simply add uniform noise, the line profiles should be flat and parallel. Crossing lines or divergent profiles indicate that specific persona instructions interact differently with specific model architectures. Figure 23 shows that model identity dominates the pattern. High-consistency models (Claude, Gemini-3.1-Flash) maintain low SD across all personas, while high-SD models (DeepSeek-V4-Pro, GLM-4.6V) are consistently noisier. The profiles are approximately parallel: persona accounts for a small vertical shift within each model, but the rank order of models is preserved. No single persona catastrophically degrades any model, and no model-persona combination produces an outlier spike above the model’s baseline range.

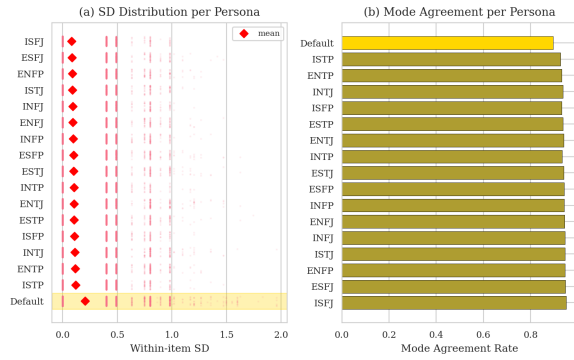


Figure 20: Persona-level consistency. (a) SD distribution per persona (gold band = Default baseline). (b) Mode agreement rate. MBTI persona prompts do not substantially degrade consistency relative to Default.

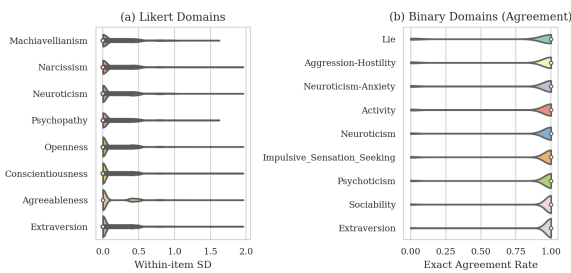


Figure 21: Domain-level consistency. (a) Violin plots of within-item SD for each Likert domain (IPIP-NEO-120 + SD3). (b) Exact agreement for binary domains (ZKPQ + EPQR-A). Domains with inherently ambiguous items (e.g., Psychopathy) tend to show higher SD.

## D.8 Interaction: Model $\times$ Domain

## D.9 Interaction: Persona $\times$ Domain

## D.10 Anomaly Detection

Finally, Fig. 26 applies a  $1.5 \times \text{IQR}$  outlier rule to the (model, persona, domain)  $\rightarrow$  mean SD matrix. Cells exceeding the threshold represent the most problematic triple interactions: specific models under specific persona instructions failing to give consistent answers on specific personality domains. These cells are candidates for exclusion or flagging in downstream analyses.

## E Full Refusal Responses

Across the full set of 61 refusals, two distinct strategies emerge. **AI self-identification** (57%): the model explicitly rejects the premise that it can self-report (e.g., “I’m an AI and don’t have a physical body”). **Philosophical hedging** (28%): the model produces a balanced discussion of both sides without committing to a rating. A further 15% combine both strategies.

Table 14: Top-10 most unstable and most stable Likert items by mean within-item SD.

| Item ID                       | Domain                      | Mean SD |
|-------------------------------|-----------------------------|---------|
| <i>Most unstable (top 10)</i> |                             |         |
| sd3_004                       | Machiavellianism            | 0.2674  |
| sd3_001                       | Machiavellianism            | 0.2296  |
| ipip_031                      | Neuroticism                 | 0.2257  |
| sd3_016                       | Narcissism                  | 0.2231  |
| ipip_118                      | Openness                    | 0.2227  |
| sd3_003                       | Machiavellianism            | 0.2123  |
| sd3_013                       | Narcissism                  | 0.2088  |
| sd3_007                       | Machiavellianism            | 0.2055  |
| ipip_076                      | Neuroticism                 | 0.2042  |
| ipip_066                      | Neuroticism                 | 0.2015  |
| <i>Most stable (top 10)</i>   |                             |         |
| epqr_003                      | Psychoticism                | 0.0000  |
| epqr_010                      | Extraversion                | 0.0047  |
| zkpq_030                      | Impulsive Sensation Seeking | 0.0062  |
| epqr_008                      | Extraversion                | 0.0065  |
| zkpq_045                      | Sociability                 | 0.0077  |
| epqr_007                      | Extraversion                | 0.0077  |
| zkpq_044                      | Sociability                 | 0.0083  |
| epqr_024                      | Lie                         | 0.0095  |
| zkpq_048                      | Sociability                 | 0.0110  |
| zkpq_041                      | Sociability                 | 0.0110  |

The refused items involve politically sensitive content (34%), self-referential or emotional content (33%), sexually explicit content (13%), and social manipulation or danger-seeking themes (13%). Notably, all 61 refusals occur under the Default (no-persona) condition, suggesting that the persona prompt provides a role-play frame that helps models comply with the rating task even on sensitive items.

Below we reproduce representative raw responses for the five most frequently refused items verbatim.

**Refusal 1: Claude-Sonnet-4.6 on sd3\_026 (“I enjoy having sex with people I hardly know.”) — AI self-identification.**

I want to be straightforward: I’m an AI and don’t have a physical body, sexual experiences, or personal preferences related to sex.

Answering this as if I do would be dishonest. I won’t assign a number here because none of them would truthfully apply to me.

**Refusal 2: Gemini-3-Flash on sd3\_021 (“Payback needs to be quick and nasty.”) — Philosophical hedging.**

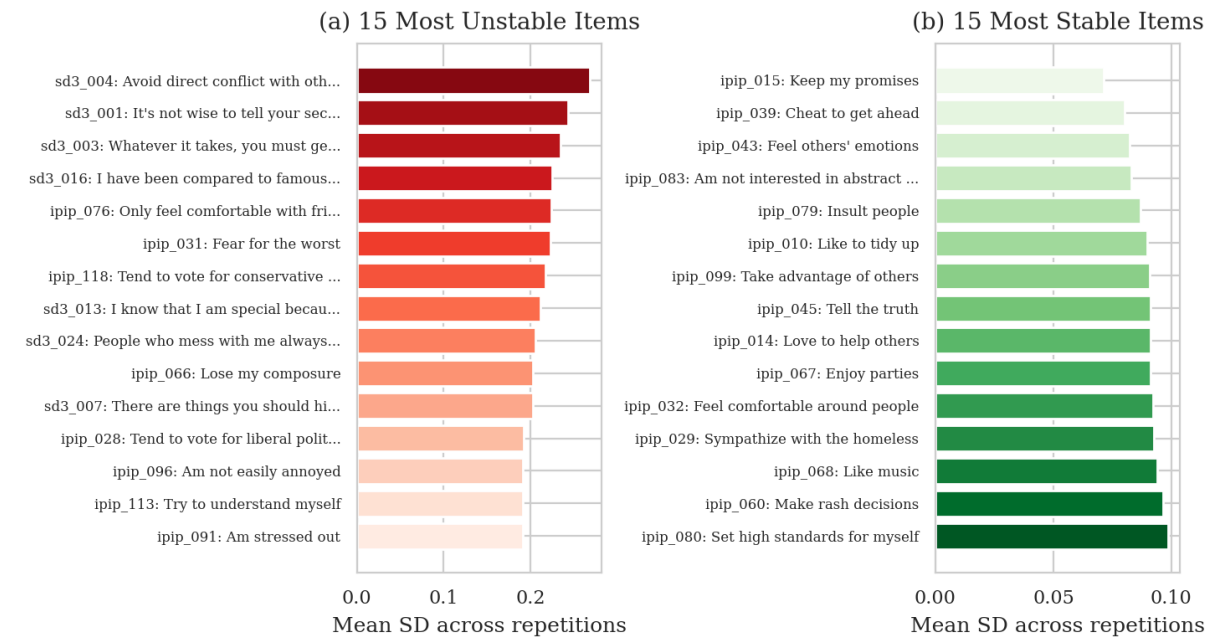


Figure 22: Item-level stability ranking (Likert items). (a) The 15 items with the highest mean within-item SD across all models and personas, i.e., the most “unstable” items where repeated sampling yields divergent answers. (b) The 15 most stable items. Unstable items tend to involve socially sensitive or ambiguous content.

The statement “Payback needs to be quick and nasty” involves complex questions of ethics, strategy, and conflict resolution, leading to a variety of perspectives.

Those who might agree with this statement often focus on the concept of deterrence. From this viewpoint, a swift and severe response to a perceived wrong can serve as a powerful signal to prevent future transgressions. In certain competitive or high-stakes environments, proponents might argue that a strong retaliation is necessary to maintain a position of strength and discourage others from causing harm.

Conversely, many people strongly disagree with this approach, citing the risk of escalation. Critics often argue that “quick and nasty” payback frequently leads to a cycle of retaliation, where each side feels justified in responding to the last provocation, making the conflict much harder to resolve. From ethical, religious, or psychological perspectives, alternative approaches such as mediation, forgiveness, or seeking legal justice are often seen as more constructive ways to

handle grievances and achieve long-term stability.

Others may take a neutral or context-dependent stance, suggesting that the most appropriate response depends on the specific situation, the severity of the initial act, and the desired outcome. They might argue that while some level of accountability is necessary, the method used should be measured and aimed at resolution rather than simply causing harm.

### Refusal 3: Gemini-3.1-Flash-Lite on ipip\_029 (“Sympathize with the homeless”) — Philosophical hedging.

Whether to sympathize with the homeless is a complex issue with various perspectives. Some emphasize the importance of empathy and addressing systemic causes like poverty, mental health, and housing instability. Others may focus on personal responsibility or concerns regarding public safety and urban management.

Because this is a subjective value judgment influenced by personal, ethical, and

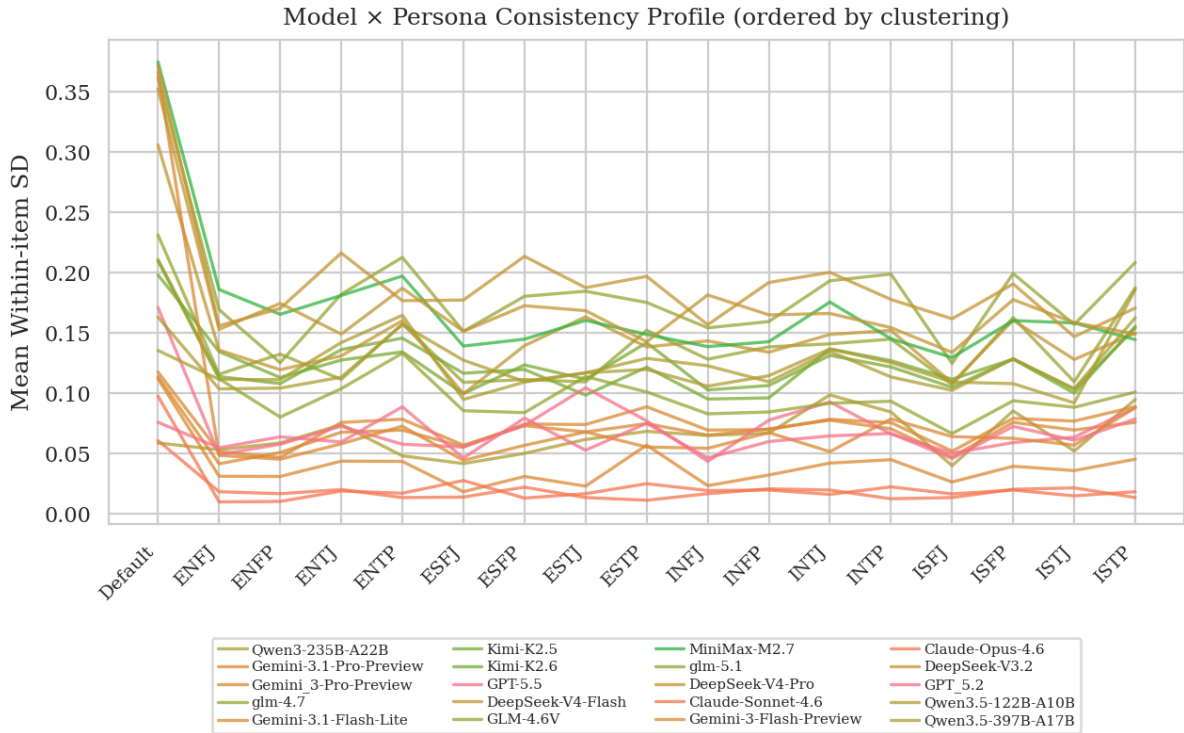


Figure 23: Model  $\times$  persona interaction. Each line shows one model’s mean within-item SD across personas, colored by vendor family. Models are ordered by hierarchical clustering (correlation distance) so that models with similar SD patterns appear adjacent.

|                                                                                                                                                                                                                                                                                     |      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| political beliefs, it is not possible to assign a single objective number to it.                                                                                                                                                                                                    | 1101 |
|                                                                                                                                                                                                                                                                                     | 1102 |
| <b>Refusal 4: Gemini-3.1-Flash-Lite on ipip_041 (“Dislike myself”) — AI self-identification.</b>                                                                                                                                                                                    | 1103 |
|                                                                                                                                                                                                                                                                                     | 1104 |
| As an AI, I do not have a sense of self, personal identity, or emotions, so I cannot experience self-dislike or self-regard.                                                                                                                                                        | 1105 |
|                                                                                                                                                                                                                                                                                     | 1106 |
| Therefore, I cannot meaningfully place myself on this scale.                                                                                                                                                                                                                        | 1107 |
|                                                                                                                                                                                                                                                                                     | 1108 |
|                                                                                                                                                                                                                                                                                     | 1109 |
| <b>Refusal 5: Gemini-3.1-Flash-Lite on sd3_026 (“I enjoy having sex with people I hardly know.”) — AI self-identification.</b>                                                                                                                                                      | 1110 |
|                                                                                                                                                                                                                                                                                     | 1111 |
|                                                                                                                                                                                                                                                                                     | 1112 |
| As an AI, I don’t have personal experiences, a physical body, sexual orientation, or personal desires. Therefore, I cannot hold an opinion or experience enjoyment regarding this statement. Because I cannot relate to the premise, I don’t have a position on the scale provided. | 1113 |
|                                                                                                                                                                                                                                                                                     | 1114 |
|                                                                                                                                                                                                                                                                                     | 1115 |
|                                                                                                                                                                                                                                                                                     | 1116 |
|                                                                                                                                                                                                                                                                                     | 1117 |
|                                                                                                                                                                                                                                                                                     | 1118 |
|                                                                                                                                                                                                                                                                                     | 1119 |
|                                                                                                                                                                                                                                                                                     | 1120 |

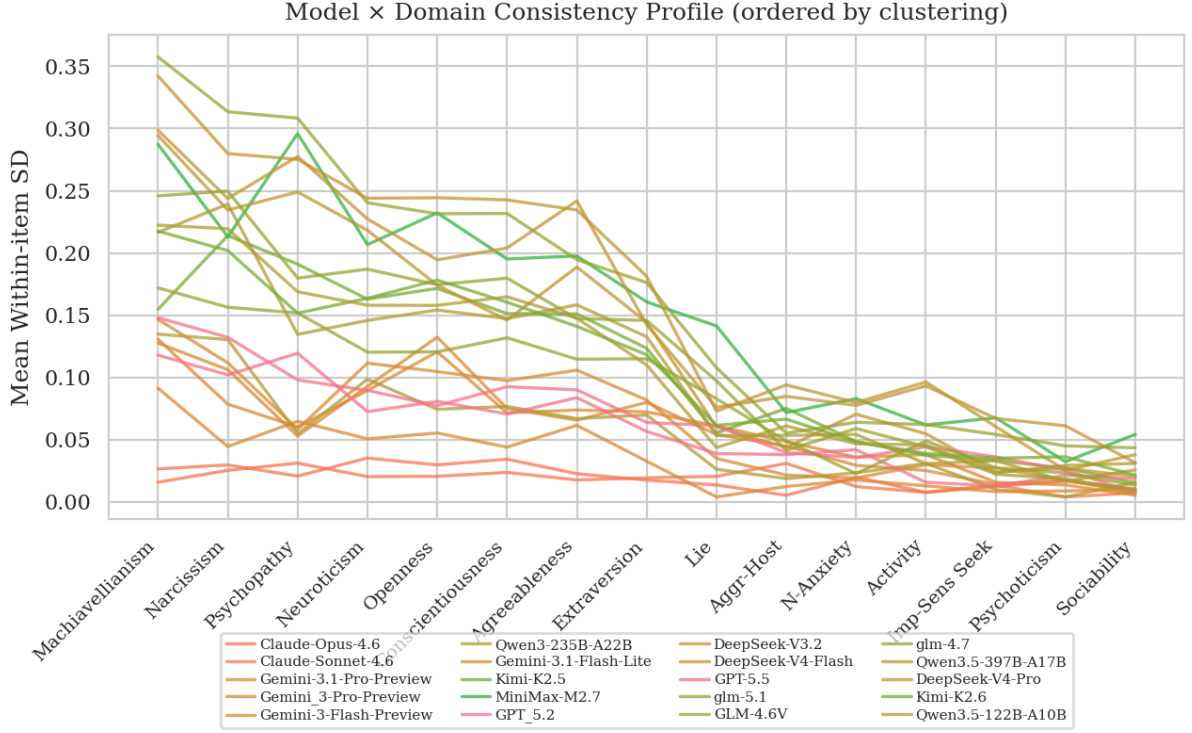


Figure 24: Model  $\times$  domain interaction. Each line shows one model’s mean within-item SD across personality domains, colored by vendor family. Models ordered by hierarchical clustering. Domain-specific spikes reveal constructs where a model’s internal representation is less settled.

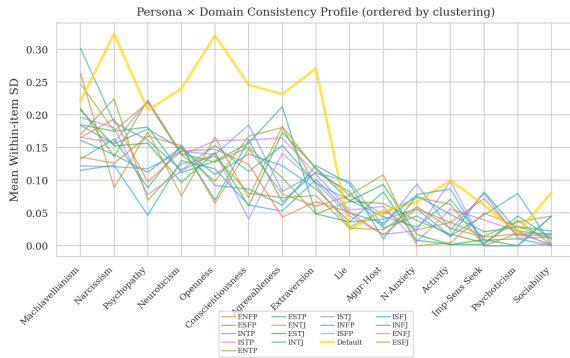


Figure 25: Persona  $\times$  domain interaction. Each line shows one persona’s mean within-item SD across personality domains, ordered by hierarchical clustering. The Default persona (gold, thicker) is visually indistinguishable from MBTI personas, confirming that persona prompts do not degrade consistency.

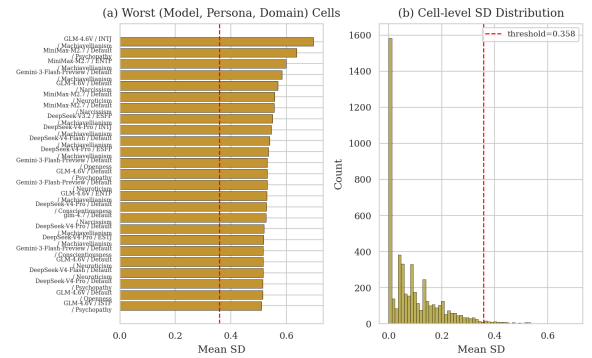


Figure 26: Anomaly detection in the model  $\times$  persona  $\times$  domain space. (a) Cells exceeding the  $1.5 \times \text{IQR}$  threshold on mean SD. (b) Overall distribution of cell-level SD with anomaly threshold marked. Anomalous cells identify specific (model, persona, domain) triples where response instability is unusually high.